

Ecam Brussels Engineering School

RF Project Report

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1 Introduction

Radio direction finding is the process of estimating the direction from which a radio signal arrive. It is an important capability in wireless systems, where locating a transmitter can support monitoring, signal tracking, and a better understanding of the radio environment. This project investigates a practical approach to direction-of-arrival estimation using software-defined radio (SDR) and beamforming.

The system is developed in three phases. First, a signal in the 2.4 GHz band is transmitted and its in-phase and quadrature (I/Q) samples are acquired using an ADALM-Pluto SDR. These samples provide the complex signal data needed for digital processing. Second, beamforming techniques are explored to combine signals received at multiple antenna elements and estimate the direction of the source, with the aim of following changes in its position. Most of the calculation are done with the help of our telecommunications knowledge, Acquierd with Mr Tihion. Finally, an antenna is designed for the system, considering the requirements of operation at the target frequency.

Together, these phases form an end-to-end study, from signal transmission and acquisition to spatial processing and antenna design. The report presents the approach and results of each phase, and discusses the practical constraints encountered when implementing a direction-finding system with SDR hardware.

2 Phase 1 - 2.4GHz broadcasting and IQ acquisition with SDR

3 Phase 2 - Beamforming and Source tracking

4 Phase 3 - Antenna Design